

Canadian Arctic Sea Ice Concentration Change Along the Northwest Passage (2015–2025)

Professional Project Report

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Executive Summary

This project analyzes changes in September sea ice concentration along the Canadian Northwest Passage between 2015 and 2025 using NOAA/NSIDC Climate Data Record Version 6. The workflow combines remote sensing data, raster analysis, vector processing, and professional cartographic design in ArcGIS Pro to evaluate changes within a 200 km analysis corridor surrounding the Northwest Passage.

1. Introduction

The Arctic is experiencing significant environmental change, making long-term monitoring of sea ice essential for navigation, environmental management, and climate research. This project compares September sea ice concentration between 2015 and 2025 to identify areas of increase and decrease along the Canadian Northwest Passage.

2. Study Area

The study area covers the Canadian Arctic Archipelago and a 200 km analysis corridor centered on the Northwest Passage. Focusing on this corridor allows the analysis to concentrate on sea ice conditions most relevant to Arctic marine navigation.

3. Data Sources

- NOAA/NSIDC Climate Data Record (CDR) Version 6 – Monthly Sea Ice Concentration (25 km NetCDF)
- National Geospatial-Intelligence Agency (NGA) – Arctic Sea Routes
- Natural Earth – Admin 0 Countries

4. Software

ArcGIS Pro

Spatial Analyst Extension

ModelBuilder (future enhancement)

5. Methodology

1. Downloaded NOAA/NSIDC monthly sea ice concentration NetCDF datasets.
2. Imported NetCDF variables into ArcGIS Pro.
3. Converted NetCDF variables to raster datasets.
4. Stored rasters in a File Geodatabase.
5. Calculated raster differences using Raster Calculator (Sealce2025 – Sealce2015).
6. Selected the Northwest Passage from the Arctic Sea Routes dataset.
7. Created a 200 km buffer around the shipping route.
8. Used Extract by Mask to isolate the analysis corridor.
9. Applied a diverging blue–white–red color scheme and produced a publication-quality layout.

6. Results

The resulting map identifies localized increases and decreases in September sea ice concentration along the Northwest Passage. The analysis demonstrates that changes are spatially variable and highlights areas that may warrant further investigation in future shipping and climate studies.

7. Skills Demonstrated

- NetCDF Processing
- Raster Analysis
- Raster Calculator
- Extract by Mask
- Buffer Analysis
- Geodatabase Management
- Coordinate Systems (EPSG:3411)
- Cartographic Design
- ArcGIS Pro

8. Limitations

Only September observations from two years were compared. The 25 km spatial resolution is appropriate for regional analysis but does not capture fine-scale local ice conditions.

9. Future Work

Extend the analysis to all years between 2015 and 2025, compare multiple seasons, incorporate AIS vessel traffic, evaluate shipping accessibility, and automate the workflow using ModelBuilder and Python.

10. Conclusion

This project demonstrates a complete GIS workflow from scientific data acquisition to spatial analysis and professional cartographic presentation. It provides a strong example of remote sensing and GIS analysis suitable for a professional portfolio.

References

NOAA/NSIDC Climate Data Record Version 6
National Geospatial-Intelligence Agency (NGA) Arctic Sea Routes
Natural Earth Admin 0 Countries